

THE BLUEPRINT & THE TOY

Intro to Classes & Objects

PYTHON PROGRAMMING | GRADE 4

PLANRS Academy

The Hook: LEGO vs. Instructions

Imagine This...

Imagine if you had a big pile of colorful LEGO blocks but absolutely no instructions.



Visual: *A pile of bright LEGO blocks next to a missing manual.*

How would you know exactly what to build? Is it a car? A castle? A robot?

In Python, a **Class** is just like that instruction manual! It tells the computer exactly how to build something step-by-step.

Without the manual (Class), the blocks are just blocks. With the manual, you can build anything!

The Blueprint

What is a Class?

A Class is a blueprint or a plan. Think of it like a Cookie Cutter used for making sweet dumplings (like Gujiya or Modak)!

 **Visual:** An Indian sweet dumpling cookie cutter.

- **Blueprint:** A plan or a shape.
- **Template:** Helps you make many things that look exactly the same.

The cutter itself isn't the sweet you eat—it's just the plan for how the sweet should look!

The Actual Item

What is an Object?

An Object is the actual thing made from the blueprint.

Think of the Maruti Suzuki Swift car parked in your society. The design at the factory is the Class, but the real car you can see and touch is the Object!

 **Visual:** A Maruti Suzuki Swift car parked on a street.

- Objects are created using the blueprint (Class).
- Each object is a real thing you can see and use.

Attributes & Methods

Has vs. Does

Every object in the world has special details and can do special things. In coding, we call these **Attributes** and **Methods**.

Attributes are what an object HAS.

Example: A Gujiya (Object) has a Shape and a Size (Small or Big).

Methods are what an object DOES.

Example: For our cookie, we have a Bake() method that makes it ready to eat!



Visual: A Hero Motocorp Bike factory where designs (Classes) become bikes (Objects).

Time to Code!

Let's Code a Student Class!

We are going to create a blueprint called Student so Python knows how to make student objects.

Step 1: The Blueprint

```
class Student:
    def __init__(self, name, grade):
        self.name = name
        self.grade = grade

    def introduce_self(self):
        print("Hi, I am", self.name)
```

Step 2: Creating an Object

Now, let's create a real student named Kabir!

```
kabir = Student("Kabir", 4)
kabir.introduce_self()
```

Kabir is an Object. He has attributes (name="Kabir", grade=4) and can do actions like `introduce_self()`!

The Blueprint Challenge

Why do we need Classes?

Classes help us keep our code neat, just like a tidy cupboard!

- Organize code logically.
- Make many objects easily from just one blueprint.

The Blueprint Challenge!

Test your knowledge:

1. Which is the Class: Cake Recipe or Chocolate Cake?
2. Name 3 attributes of a School Bus object.
3. What does a Remote Control do? Give one example method.
4. True or False: An Object is a blueprint.

Great job, Programmer! You've mastered the basics of Classes and Objects!

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Student Practice Workbook (Grade 4)

Test your programming powers! Answer these questions based on what you learned about blueprints and objects.

1. What everyday toys are used in the lesson to explain the difference between parts and an instruction manual?

.....

2. Fill in the blank: In Python, a _____ works exactly like an instruction manual or a blueprint.

.....

3. If you use a Modak or Gujiya cookie cutter to shape sweets, is the cookie cutter itself considered the Class or the Object?

.....

4. What is an Object in Python? Describe it using a real-world example from the lesson.

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5. If the design plans at a factory represent a Class, what does a finished Maruti Suzuki Swift car parked outside represent?

.....

6. Explain what an "Attribute" means in coding. Give an example of an attribute for a Gujiya.

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7. What is a "Method" in Python? What does a method represent an object doing?

.....

8. If you have a computer game with a dog character, would "bark()" be an attribute or a method? Why?

.....

9. Which keyword do we use in Python to start writing a new blueprint code block?

.....

10. In the coding example, when we wrote `kabir = Student("Kabir", 4)`, what attributes were given to Kabir?

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11. What action does the method `introduce_self()` print to the screen in our code?

12. Give two main reasons why programmers use Classes instead of writing loose code everywhere.

13. True or False: You can make many different individual objects using only one single Class blueprint.

14. If a "Cake Recipe" is a Class, what would the actual "Chocolate Cake" sitting on the kitchen counter be?

15. Think of a Bicycle object. Name one attribute it HAS and one method it DOES.
